Exact Early Filters and Search-Pipeline Roadmap for the Monohedral Disk Solver

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Abstract

This note reorganizes the current filtering strategy around exact invariants that can be tested substantially earlier than contour-word resolution. It also records an explicit source audit concerning exterior intersections: throughout this note, a tile is *not* assumed to meet the outer circle. The only safe statement used is that an exterior intersection, when nonempty, is a single closed connected circular arc (possibly a singleton). Empty exterior intersections are allowed and occur in known monohedral disk tilings.

The first main result is a graph-level perimeter regularizability criterion: before any contour offsets are chosen, the positive-length contact structure must admit strictly positive interface weights giving every tile the same weighted degree. After collapsing parallel contacts and replacing positive-length exterior contact by a loop, this is exactly positive graph regularizability, equivalently total support of the symmetric adjacency pattern. A second, independent early constraint comes from the exact relative isoperimetric problem in a Euclidean disk: since every tile has area $\pi R^2/n$, the total length of its internal interfaces has an explicit lower bound proportional to the circumference $2\pi R$. This produces additional linear inequalities on the same graph-level interface variables, still before offsets.

For Stein configurations with an imposed geometric rotation, another new result gives a strong orbit obstruction: every nondegenerate exterior arc has angular length strictly less than one rotational period. Hence at least two complete peripheral tile orbits are required, so an order-q rotational Stein configuration has at least $1 + 2q$ tiles. A further theorem generalizes the existing cyclic-hub obstruction: once the length system proves that any connected central interface has length equal to an integral number of rotational contour periods, the configuration is impossible by periodic signed-curvature cancellation.

The second half maps these results onto the existing implementation. It proposes a staged pipeline beginning with graph-only support tests, the relative-isoperimetric perimeter LP and symmetry filters, then prefix-level positive-length feasibility, exterior-arc classes, the existing radius-R topology propagation, a productionized nested linear curriculum, specialized periodic word-equation handlers, and only then the generic Nielsen–Levi and geometric solvers. Heuristic scores remain strictly separate from exact rejection.

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1 Scope and conventions

This document assumes familiarity with the current solver and therefore does not restate the full pipeline. The relevant existing components are referred to by their current names, in particular the length oracles (`LengthFeasibilityOracle` and `StrictLengthFeasibilityOracle`), the hybrid linear oracle (`HybridMarginOracle`), the prefix and complete radius-arc topology filters, and `PrewordLinearInvariantFilter` / `PrewordPruningPipeline`.

The geometric setting is a finite monohedral tiling

$$B_R = D_1 \cup \cdots \cup D_n$$

of the closed Euclidean disk by congruent Jordan regions with pairwise disjoint interiors. Whenever perimeter or regular signed curvature is used, rectifiability or the current piecewise-C^2 hypothesis is stated explicitly. A *positive-length interface* means one of the interface components represented by the planar map, not a mere point contact.

A *Stein configuration* means that the center O of B_R lies in the interior of one tile. This tile is called the *center tile*. Since its interior contains a neighborhood of O, no other tile contains O.

The note distinguishes three statuses:

- **Established**: already proved in the current documentation or in the cited literature.
- **New theorem**: proved below from the stated hypotheses.
- **Engineering proposal**: a sound use or reordering of existing necessary conditions, but not a new mathematical theorem.

2 Existing exact structure worth preserving

2.1 Exterior arcs: source audit and safe formulation

This point is important because an earlier reading of the literature incorrectly treated boundary contact as compulsory. It is not. Haddley and Worsley explicitly exhibit monohedral disk tilings in which some tiles do not intersect the bounding circle at all; for example, in their D_n^t families with $n \geq 5$, half of the tiles meet the boundary and half do not [3]. Therefore no filter in this note may infer

$$D_i \cap \partial B_R \neq \varnothing$$

from congruence alone.

The safe geometric statement extracted from the argument of Kurusa, Lángi and Vígh is the following.

Proposition 2.1 (Safe exterior-intersection statement)**.** *For every tile D_i, the set*

$$S_i = D_i \cap \partial B_R$$

is either empty or a closed connected circular arc; a nonempty degenerate arc may be a singleton. In particular, if a tile has positive-length exterior contact, that contact is one connected circular arc, not several separated arcs.

Proof. If S_i is empty or a singleton, the assertion is immediate. If it contains at least two points, apply the proof of Lemma 2.6 of Kurusa–Lángi–Vígh to that tile: choose two farthest points of the compact set S_i and run their circumdisk/spindle argument. The proof shows that the shorter circle arc between the extremal points is contained in the tile, hence the whole exterior intersection is connected. The only correction needed for solver use is that emptiness must be allowed explicitly; nonemptiness is not a consequence of the lemma. [1] □

For nonempty exterior arcs, the second structural statement of Kurusa–Lángi–Vígh remains applicable: after transporting two such arcs to a common prototile, their relative interiors are disjoint unless the arcs coincide. Empty sets play no role in that assertion. The current file `six_structural_results.tex` should therefore be read with "empty allowed" made explicit whenever it speaks of exterior arcs.

The current `RadiusArcTopologyFilter` already uses only local facts attached to *actual* exterior arcs: a radius-R subarc transported across an interface reverses convex/concave sign; a smooth radius-R arc cannot cross a hard endpoint; and a single atomic open interval cannot simultaneously carry both signs. These uses are safe because the filter seeds only from exterior interfaces that are present in the compiled case.

Audit of the new results below. The nondegenerate repetition theorem first proves that nondegenerate exterior arcs exist and never counts empty intersections as repetitions. The rotational Stein theorem uses Theorem 2.1 only conditionally: if the rotation-fixed center tile met the outer circle, its nonempty intersection would have to be one connected invariant arc; this leads to a contradiction, so the conclusion is that the intersection is empty. The rotational arc bound then starts from a specifically chosen nondegenerate physical exterior arc. Thus none of the later arguments uses the false implication "every tile meets the boundary."

2.2 Stein center tile forces radius-R concavity

The current documentation proves the following.

Theorem 2.2 (Existing: strict radius-R concavity)**.** *If $O \in \operatorname{int} D_c$ in a nontrivial monohedral tiling of B_R, then the prototile boundary contains a nondegenerate radius-R circular arc that is concave relative to the prototile.*

This is already represented by `requires_radius_r_concavity` and by the `radius_concavity` stage in `PrewordLinearInvariantFilter`. It should remain a necessary-condition constraint, not a graph-only rejection.

2.3 Current cyclic-hub obstruction

The current `cyclic_hub_obstruction.tex` proves, under the piecewise-C^2 hypothesis, that a rotationally fixed center tile cannot have its complete boundary partitioned into one orbit of q connected interfaces with one orbit of q congruent neighbors. The proof is by periodic regular signed curvature and cancellation across the shared interface. Arbitrary outer rings do not repair the contradiction.

This theorem is correct but its hypotheses are narrower than necessary. Section 7 gives a direct strengthening that can be triggered from the length row space.

2.4 The present preword linear curriculum

The current `PrewordLinearInvariantFilter` already contains the nested stages

> `positive_lengths`, `radius_local`, `smooth_turn_local`, `radius_global`, `smooth_turn_global`, `radius_concavity`, `isoperimetric`, `point_turns`.

Every production rejection is ultimately certified exactly through rational arithmetic. This is the right contract. The main implementation issue is not mathematical soundness but how early the weaker prefixes of these systems are exposed to the search tree.

3 New graph-only result: perimeter regularizability

This is the most immediate new filter because it needs no contour offsets, no weak order, no word equation, and no geometric coordinates.

3.1 The perimeter system

Assume that tile boundaries are rectifiable and that the planar map lists all positive-length interface components. Give every physical interface component e a variable $x_e > 0$ equal to its arclength. For an internal interface, the same x_e belongs to both incident tile boundaries. For an exterior interface, x_e belongs only to its tile.

Because all tiles are congruent, they have a common perimeter $P > 0$. Therefore every tile v satisfies

$$\sum_{e \ni v} x_e = P. \tag{3.1}$$

After division by P, this is a strictly positive linear feasibility problem depending only on the positive-length incidence structure.

3.2 Loop-augmented contact graph

Collapse parallel positive-length contacts between the same two tiles to one support edge. If a tile has at least one positive-length exterior interface, add one loop at that tile. Denote the resulting undirected graph with loops by H.

The weight of a collapsed internal edge is the sum of the physical lengths of all parallel interfaces. The weight of a loop is the total length of all positive exterior interfaces of that tile. Conversely, every positive aggregate weight can be split into arbitrarily many positive summands, so this collapse loses no information for mere perimeter feasibility.

Theorem 3.1 (Perimeter regularizability criterion)**.** *For a fixed positive-length contact map with rectifiable tile boundaries, the perimeter equations* admit strictly positive lengths for every represented physical interface *if and only if the loop-augmented support graph H admits strictly positive edge/loop weights for which every vertex has the same weighted degree.*

Proof. Suppose first that the physical interface lengths exist. For every pair of tiles, sum the lengths of all parallel internal interfaces and use the sum as the weight of their support edge in H. For every tile incident with the exterior, sum the lengths of all its exterior interfaces

and use the sum as the loop weight. All aggregate weights are strictly positive. Equation (3.1) says exactly that the weighted degree of every vertex is P.

Conversely, suppose H has strictly positive regularizing weights with common weighted degree P. If a support edge represents $r \geq 1$ physical interfaces, split its positive weight into r arbitrary positive summands. Do the same for a loop representing several exterior interfaces. The sum incident with every tile remains P, so the resulting physical interface lengths satisfy (3.1) and are all strictly positive. □

Thus the relevant graph-theoretic notion is exactly *positive regularizability*. For a symmetric adjacency pattern, positive regularizability is equivalent to *total support*: every nonzero adjacency entry lies in a supported permutation matrix [2]. In undirected language this becomes the following.

Corollary 3.2 (Complete graph-only characterization)**.** *The perimeter necessary condition holds if and only if every support edge and every support loop of H belongs to an undirected spanning cycle forest: a spanning subgraph whose connected components are single edges, cycles, or loops.*

This is an exact characterization of the *perimeter relaxation*; it is not a sufficient condition for a tiling.

3.3 A very small matching algorithm

Let A be the symmetric 0–1 adjacency matrix of H, including diagonal entries for loops. Form a bipartite graph $B(H)$ with a left and a right copy of the tile vertices and an edge $i_L j_R$ whenever $A_{ij} = 1$.

Total support of A is equivalent to the statement that every edge of $B(H)$ belongs to at least one perfect matching. Hence, for the solver's sizes, the following deliberately simple algorithm is already sufficient:

1. Check that $B(H)$ has a perfect matching. If not, reject.
2. For each bipartite support edge $i_L j_R$, force it and check whether the remaining bipartite graph has a perfect matching. If any support edge fails, reject.

With fewer than twenty tiles this is negligible even with a generic matching implementation. If desired, all allowed edges can instead be classified from one perfect matching by the usual alternating-cycle decomposition.

The important point is architectural: this entire test belongs immediately after the planar map is known. It should not wait for `LengthFeasibilityOracle`, because the latter only appears after a contour placement has already generated atomic length rows.

3.4 Cheap Hall-type certificates

The complete total-support test is already inexpensive, but several $O(n + m)$ or small-subset tests can precede it.

Call a tile *perimetrically interior* if it has no positive-length exterior interface, i.e. its vertex has no loop in H.

Proposition 3.3 (Independent-set perimeter obstruction)**.** *Let S be an independent set of perimetrically interior tiles, and let $N(S)$ be its set of tile neighbors. Then every monohedral realization satisfies*

$$|N(S)| \geq |S|.$$

If equality holds, every tile in $N(S)$ has no positive-length interface except interfaces to tiles in S.

Proof. All boundary length of a tile in S lies on interfaces from S to $N(S)$, because S is independent and has no exterior interfaces. Summing the perimeter equations over S gives total cross-interface weight $|S|P$.

The same cross interfaces contribute to the perimeter sums of the vertices in $N(S)$. Their total available perimeter is $|N(S)|P$, and all interface weights are nonnegative, hence $|S|P \leq |N(S)|P$.

If equality holds, the cross interfaces already consume the complete perimeter sum of $N(S)$. Since every represented interface has strictly positive length, no vertex in $N(S)$ can have any additional incident support edge or loop. □

Corollary 3.4 (Bipartite-region deficit)**.** *Let U be a set of perimetrically interior tiles whose induced contact graph is bipartite with parts A and B. Then A has at least $|A| - |B|$ distinct neighbors outside U whenever $|A| > |B|$, and symmetrically for B.*

Proof. The set A is independent. Its neighborhood is the disjoint union of B and its neighbors outside U. Apply Theorem 3.3. □

These are useful not because the matching test is expensive, but because they produce tiny human-readable rejection witnesses and can be evaluated while a graph is still being constructed.

4 Established relative-isoperimetric filter at map level

The perimeter regularizability criterion uses congruence but not the fact that the ambient region is a disk of a known area. The classical relative isoperimetric problem in a Euclidean ball supplies a second exact constraint on the same interface-length variables.

4.1 Relative perimeter of one tile

For a rectifiable tile $D_i \subset B_R$, let

$$I_i = \mathcal{H}^1(\partial D_i \cap \operatorname{int} B_R)$$

be its *relative perimeter*: the length of the tile boundary lying strictly inside the ambient disk. Under the planar-map hypothesis used by the solver, this is exactly the sum of the positive-length internal interfaces incident with D_i, up to sets of one-dimensional measure zero. Let

$$T = \mathcal{H}^1(\partial B_R) = 2\pi R$$

be the circumference of the ambient disk. Since the tiles are congruent and cover the disk, every tile has area

$$|D_i| = \frac{\pi R^2}{n}.$$

The isoperimetric problem inside a Euclidean ball is classical: for a prescribed area not exceeding half the disk, the least possible relative perimeter is attained by a circular cap whose free circular arc meets the boundary of the ambient disk orthogonally; at half area the limiting minimizer is a diameter. This is the planar case of the classification of isoperimetric hypersurfaces in a Euclidean ball by Bokowski–Sperner and Almgren; see also the disk-partition literature for the same orthogonality condition [4, 5].

To make the bound directly usable, parameterize the minimizing cap by the half-angle $\theta \in (0, \pi/2]$ subtended by its outer-circle contact. Orthogonality gives an auxiliary-circle radius $R \tan\theta$. The cap area and the length of its free internal arc are

$$A(\theta) = R^2 \left[\theta + \tan^2\theta \left(\frac{\pi}{2} - \theta \right) - \tan\theta \right], \tag{4.1}$$

$$I(\theta) = 2R \tan\theta \left(\frac{\pi}{2} - \theta \right). \tag{4.2}$$

The first quantity increases continuously from 0 to $\pi R^2/2$, so for every $n \geq 2$ there is a unique $\theta_n \in (0, \pi/2]$ satisfying

$$\theta_n + \tan^2\theta_n \left(\frac{\pi}{2} - \theta_n \right) - \tan\theta_n = \frac{\pi}{n}. \tag{4.3}$$

For $n = 2$, the equation is understood in the limiting sense $\theta_2 = \pi/2$.

Theorem 4.1 (Relative-isoperimetric interface bound)**.** *Assume rectifiable tile boundaries. Define*

$$\lambda_n = \frac{\tan\theta_n \left(\frac{\pi}{2} - \theta_n \right)}{\pi}.$$

Then every tile in a monohedral tiling of B_R by $n \geq 2$ tiles satisfies

$$I_i \geq \lambda_n T. \tag{4.4}$$

This remains true for a tile having empty exterior intersection.

Proof. The tile has area $\pi R^2/n$. By the relative isoperimetric theorem in the Euclidean disk, its relative perimeter is at least the free-boundary length of the minimizing orthogonal circular cap of the same area. Equations (4.2)–(4.3) give

$$I_i \geq 2R \tan\theta_n \left(\frac{\pi}{2} - \theta_n \right) = \lambda_n (2\pi R) = \lambda_n T.$$

The relative isoperimetric problem allows competitors with arbitrary contact with the container boundary, including none, so no boundary-contact assumption is used. □

4.2 Linear form for the existing perimeter variables

Let $x_e > 0$ denote the physical interface variables from Section 3. Write E_{out} for exterior interfaces and $E_{\text{int}}(v)$ for internal interfaces incident with tile v. Then

$$T = \sum_{e \in E_{\text{out}}} x_e, \qquad I_v = \sum_{e \in E_{\text{int}}(v)} x_e.$$

Thus Theorem 4.1 is exactly the family of linear inequalities

$$\sum_{e \in E_{\text{int}}(v)} x_e \;\geq\; \lambda_n \sum_{e \in E_{\text{out}}} x_e \qquad \text{for every tile } v. \tag{4.5}$$

These inequalities can be added to the graph-level perimeter system before any contour offset is selected. Parallel contacts may still be collapsed to aggregate support weights because only sums occur.

The constants λ_n are not generally rational. Production rejection should therefore use a precomputed certified rational lower bound

$$0 < \underline{\lambda}_n \le \lambda_n$$

and enforce the weaker rational inequality with $\underline{\lambda}_n$. This is sound: replacing the exact lower bound by a smaller rational number can only lose pruning. Since the solver uses only the finite set of tile counts of interest, these constants can be certified offline by directed-rounding interval arithmetic and then stored as exact fractions.

For orientation, the approximate values relevant to the current search are:

n	λ_n	n	λ_n	n	λ_n
3	0.301412	7	0.228294	11	0.189740
4	0.278547	8	0.216512	12	0.182812
5	0.258712	9	0.206369	13	0.176598
6	0.242166	10	0.197529	18	0.152968
				19	0.149298

These decimal values are explanatory only and must not be used as formal certificates without directed rounding.

4.3 Relation to the existing global isoperimetric row

Summing (4.4) over all tiles gives

$$\sum_i I_i \ge n\lambda_n T.$$

Since the sum of all tile perimeters is 2 times the total internal-interface length plus T, this yields a global scalar inequality. For small n it can be slightly stronger than the current global isoperimetric row, but for larger n it is not. The main value of (4.5) is therefore *local*: every individual tile must reserve enough internal boundary length, and this can conflict with the distribution of exterior weights even when all global perimeter sums remain feasible.

Algorithmically, the recommended map-level cascade is therefore:

1. pure support/total-support regularizability, requiring no numerical constants;
2. only for survivors, the tiny rational `RelativePerimeterFlowLP` consisting of equal-perimeter equations, positivity, and the certified inequalities (4.5).

This is a useful fragmentation because the matching test is essentially free, while the second stage injects genuine disk geometry without waiting for contour positions.

5 New strengthening of exterior-arc repetition

The current structural note proves that among all transported exterior arcs, allowing empty and singleton arcs, some pair coincides. For search purposes the nondegenerate version is more useful.

Theorem 5.1 (Nondegenerate exterior-arc repetition)**.** *Let $D_1, \dots, D_n$ be a nontrivial monohedral tiling of B_R by Jordan regions. Transport every set $S_i = D_i \cap \partial B_R$ to a common prototile by arbitrary fixed tile congruences. Then at least two* nondegenerate *transported exterior arcs coincide.*

Proof. By Theorem 2.1, every nonempty S_i is a closed connected circular arc. For nonempty transported arcs, Kurusa–Lángi–Vígh prove that two such arcs either have disjoint relative interiors or coincide [1].

There must be at least two nondegenerate physical exterior arcs. Indeed, if exactly one S_i were nondegenerate, the union of the finitely many S_i would force that single arc to be the whole circle. A Jordan region contained in B_R and containing the complete boundary circle must equal B_R, contradicting $n > 1$.

Assume that no two nondegenerate transported arcs coincide. Their relative interiors are then pairwise disjoint. The physical exterior arcs cover ∂B_R and overlap only on zero-length sets, so the sum of their arclengths is $2\pi R$. Transport preserves arclength. Every transported exterior arc lies on $\partial \operatorname{conv} D_1$ and has radius R, so these arcs contribute total turning 2π to the boundary of the convex hull.

The total turning of a convex Jordan curve is 2π. Hence every remaining part of $\partial \operatorname{conv} D_1$ has zero turning and is a straight segment tangent to the adjacent radius-R arcs. The standard parallel-body argument used in Lemma 2.9 of [1] gives

$$\operatorname{conv} D_1 = P + B_R$$

for a convex polygon P, possibly a point. Since $D_1 \subseteq B_R$, its circumradius is at most R. But the circumradius of $P + B_R$ equals R only when P is a point. Thus $\operatorname{conv} D_1 = B_R$.

Every point of ∂B_R is then an extreme point of $\operatorname{conv} D_1$ and therefore belongs to D_1. A Jordan region contained in B_R and containing ∂B_R is the whole disk, again contradicting $n > 1$. Hence two nondegenerate transported arcs coincide. □

Algorithmic consequence. Before enumerating a general weak order, the positive-length exterior arcs may be treated as a family of prototype intervals satisfying two exact combinatorial rules:

1. two classes either coincide exactly or have disjoint relative interiors;
2. at least one coincidence class has multiplicity at least two.

This suggests enumerating exterior-arc coincidence classes and their cyclic order before inserting the remaining contour occurrences.

6 New Stein rotation theorem: one peripheral orbit is never enough

This section gives a graph-family rejection that is substantially stronger than the previous arithmetic orbit test.

Lemma 6.1 (Only the center tile can have a nontrivial rotational stabilizer)**.** *Let a nontrivial rotation r about O preserve a monohedral tiling of B_R. If a Jordan tile D is setwise fixed by a nontrivial power of r, then $O \in \operatorname{int} D$.*

Proof. Let s be a nontrivial power of r fixing D. The restriction $s|_D$ is a continuous self-map of a topological closed disk, so Brouwer's fixed-point theorem gives a fixed point in D. The ambient nontrivial rotation s has the unique fixed point O, hence $O \in D$.

Suppose $O \in \partial D$. The restriction of s to the Jordan curve ∂D is an orientation-preserving finite-order homeomorphism of a topological circle with a fixed point. Such a homeomorphism is the identity: after cutting the circle at the fixed point, it becomes an increasing finite-order homeomorphism of an interval, and any non-fixed point would move monotonically under iteration and could not return after finitely many steps. Thus every point of ∂D would be fixed by the ambient rotation, impossible. Therefore $O \in \operatorname{int} D$. □

In a Stein configuration the center tile is therefore fixed, and every other tile belongs to a full orbit of size q under an order-q rotation.

Lemma 6.2 (The fixed center tile has no exterior arc)**.** *In a nontrivial order-q rotational Stein configuration, the center tile has empty intersection with ∂B_R.*

Proof. If its exterior intersection is nonempty, Theorem 2.1 says that it is a connected circular arc; it is also invariant under the nontrivial rotation. A singleton on ∂B_R cannot be invariant because the rotation has no fixed point on the outer circle. A proper nondegenerate connected arc cannot be invariant either. If the intersection were the whole circle, the center tile would equal B_R. Hence the intersection is empty. □

Theorem 6.3 (Rotational exterior-arc bound)**.** *Consider a nontrivial Stein tiling of B_R invariant under a geometric rotation of order $q \geq 2$. If S is any nondegenerate physical exterior arc of any tile, and θ is its central angle on ∂B_R, then*

$$\theta < \frac{2\pi}{q}.$$

Consequently the positive-length exterior tiles cannot form a single orbit of size q.

Proof. Let D_0 be the fixed center tile and let D be a tile carrying the exterior arc S. Choose an isometry $g : D \to D_0$ and put $J = g(S)$. As in Remark 2.7 of [1], J is a radius-R circular arc contained in $\partial \operatorname{conv} D_0$.

For $j = 0, \ldots, q-1$, the arc $r^j(J)$ is again a radius-R arc on $\partial \operatorname{conv} D_0$ with the same angle θ. Their relative interiors are pairwise disjoint. Indeed, if J and $r^d(J)$ had a nondegenerate overlap for some $0 < d < q$, their supporting radius-R circles would coincide. The center of that supporting circle would then be fixed by the nontrivial rotation r^d, so it would be O. Hence J would lie on ∂B_R, giving a nondegenerate exterior arc of D_0, contradicting Theorem 6.2.

The q arcs therefore contribute total convex-boundary turning $q\theta \leq 2\pi$. Suppose equality held. Then these radius-R arcs already exhaust the total turning of $\partial \operatorname{conv} D_0$. Exactly as in the proof of Lemma 2.9 of [1], the remaining convex-hull boundary consists of tangent line segments and

$$\operatorname{conv} D_0 = P + B_R$$

for a convex polygon P. Since $D_0 \subseteq B_R$, the same circumradius argument forces P to be a point and $D_0 = B_R$, impossible in a nontrivial tiling. Hence $q\theta < 2\pi$.

Finally, one tile orbit of size q contributes q congruent exterior arcs, all having the same angle by rotational symmetry. Their total angular measure is strictly less than 2π, so one orbit cannot cover the outer circle. □

Corollary 6.4 (Two-orbit and tile-count obstruction)**.** *In an order-q rotational Stein configuration:*

1. *every noncentral tile orbit has size q;*
2. *at least two noncentral tile orbits contain positive-length exterior arcs;*
3. *the total number n of tiles satisfies*

$$n = 1 + kq \quad \text{with } k \geq 2,$$

so in particular $n \geq 1 + 2q$.

Proof. The orbit statement follows from Theorem 6.1. The outer circle has positive length and is covered by the finitely many exterior intersections, so at least one nondegenerate exterior orbit exists. By Theorem 6.3, one such orbit cannot cover the whole circle, hence at least two are needed. Counting the fixed center tile and full noncentral orbits gives the formula. □

Corollary 6.5 (Immediate consequences in the present search range)**.** *For a Stein search:*

1. *no configuration with at most three tiles exists, by Corollary 1.3 of [1];*
2. *with 18 tiles, no nontrivial rotational symmetry is possible;*
3. *with 19 tiles, the only possible rotational orders are $2, 3, 6, 9$.*

Proof. The first item is published in [1]. For the others, Theorem 6.4 implies that q divides $n - 1$ and $q \leq (n - 1)/2$. For $n = 18$, q would divide 17, so the only nontrivial possibility is $q = 17$, which violates $q \leq 8.5$. For $n = 19$, q divides 18 and $q \leq 9$, giving $2, 3, 6, 9$. □

Important scope. This theorem requires an actual geometric rotation of the tiling, not merely an abstract automorphism of the contact graph. It is therefore applicable after the solver has selected a geometric symmetry action on the planar map.

7 New generalization of the cyclic-hub obstruction

The existing cyclic-hub theorem gets its contradiction because one hub interface has length exactly one rotational period of the central contour. The complete one-orbit covering hypothesis is merely one way to force that length equality.

Theorem 7.1 (Period-length central-interface obstruction)**.** *Assume piecewise-C^2 tile boundaries. Let a monohedral disk tiling be invariant under a rotation of order $q \geq 2$, and let a tile D_0 be fixed by that rotation. Write $P = \mathrm{per}(D_0)$. Suppose a connected positive-length interface $C \subset \partial D_0$ with another tile has length*

$$\mathrm{len}(C) = \frac{mP}{q}$$

for some integer $m \geq 1$. Then the tiling is impossible.

Proof. Set $L = P/q$. Rotational invariance makes the regular signed-curvature density of the positively oriented boundary of D_0 L-periodic in arclength. Let K be the regular signed curvature on any boundary interval of length L. Every interval of length mL then has regular signed curvature mK.

The interface C, viewed as part of the positively oriented boundary of D_0, therefore contributes mK. Viewed from the neighboring tile, the same physical regular arc is traversed in the opposite direction and contributes $-mK$.

Transport the neighboring tile to the common prototile D_0 by a congruence. Signed regular curvature in positive boundary orientation is preserved under this transport; for an orientation-reversing isometry, reversal of the plane orientation and reversal of boundary orientation compensate. The image of C is a connected boundary interval of D_0 of the same length mL and regular signed curvature $-mK$.

Periodicity of D_0 says that this image interval must instead have regular signed curvature mK. Thus $mK = -mK$, and since $m > 0$, $K = 0$. The full regular signed curvature of every tile is therefore $qK = 0$.

On the other hand, summing regular signed curvature over all tile boundaries cancels every smooth internal interface and leaves the positively oriented outer circle, whose regular signed curvature is 2π. This contradiction proves the theorem. □

Corollary 7.2 (Current cyclic hub as a special case)**.** *If the fixed tile boundary is partitioned into one orbit of q congruent connected interfaces, each interface has length P/q and Theorem 7.1 applies with $m = 1$.*

Algorithmic certificate. Normalize the atomic contour lengths so that their sum is one. For every connected central interface whose contour path is already fixed, let ℓ_C be its length row. For every integer $m = 1, \dots, q$, test whether the current exact length row space implies

$$q\ell_C = m.$$

If it does, reject immediately. The current code already contains exact row-space reduction machinery, so this is a very small addition. The same test can be run on a weak-order prefix as soon as the interface path and the necessary length equations are stable.

8 Where these theorems enter the current program

The main recommendation is to move from a small number of relatively monolithic stages to a monotone ladder of increasingly expensive necessary conditions. Each stage should consume only information already fixed by the search prefix.

8.1 Stage A: planar map and imposed symmetry only

This stage should contain no contour offsets.

A1. Existing combinatorial validity. Keep planarity, the current degree constraints, map consistency, and the existing symmetry-action validity checks.

A2. New `PerimeterRegularizabilityFilter`. Build the loop-augmented support graph H from the tile–tile positive-length contacts and the tiles with positive-length `OUTSIDE` contact. Apply, in increasing cost order:

1. obvious independent-set/Hall certificates, preferably while the graph itself is being generated;

2. one supported perfect-matching test for nonnegative perimeter feasibility;
3. the complete total-support test, equivalently "every support edge is allowed by some perfect matching".

A rejection here removes the complete subtree of all offset, parity and word choices for that map.

A3. New `RelativePerimeterFlowFilter`. For maps surviving total support, reuse one positive variable per aggregate internal contact and one loop variable per exterior-contacting tile. Add the equal-perimeter equations and the certified rational forms of Theorem 4.1:

$$I_v \geq \underline{\lambda}_n T \qquad (v \text{ any tile}).$$

This stage is still independent of contour offsets. It is strictly more geometric than regularizability because it uses the known tile area and the circular container, but it remains a tiny linear feasibility problem. Keep the fast-float/exact-certificate policy: a floating failure is only a trigger for an exact rational rejection using the stored rational lower bound.

A4. New `SteinRotationOrbitFilter`. When the selected symmetry is a genuine order-q geometric rotation and the search branch is Stein:

1. require exactly one fixed tile orbit, namely the center tile;
2. require every other tile orbit to have size q;
3. require at least two positive-length exterior tile orbits;
4. equivalently require $n \geq 1 + 2q$;
5. reject the whole $n \leq 3$ Stein domain by [1].

For the present special sizes, this removes every rotational branch at $n = 18$ and all orders except $2, 3, 6, 9$ at $n = 19$ before any contour search.

A5. Existing cyclic-hub map theorem. If the map plus selected geometric symmetry already certifies the exact H1–H3 hypotheses in `cyclic_hub_obstruction.tex`, reject at map level. Do not wait for the generic preword system to rediscover the contradiction.

8.2 Stage B: symmetry-reduced contour skeleton

B1. Enumerate exterior-arc coincidence classes before a general weak order. Use Theorem 5.1 and the KLV disjoint-or-coincident rule. The outer arcs are not generic contour occurrences: they are radius-R intervals that must form disjoint prototype classes, with at least one repeated nondegenerate class.

A natural outer-first skeleton is:

1. partition positive-length exterior interfaces into proposed coincidence classes compatible with copy parity and the imposed symmetry;
2. choose the cyclic order of the distinct classes on the prototype;
3. only then insert internal-interface endpoints and unresolved point occurrences.

This should reduce the weak-order branching factor substantially, especially in symmetric cases.

B2. Quotient the contour by the stabilizer of a fixed tile. If a tile is genuinely fixed by a rotation of order q, its contour is q-periodic. Do not enumerate q copies of equivalent occurrence data independently. Enumerate one fundamental sector and lift it equivariantly. Reflection-fixed branches should similarly be expressed by a fundamental half-contour plus its mirror, with special handling of the two axis-fixed boundary points.

This is not a rejection theorem; it is a search-space quotient and can yield an exponential reduction in the number of weak orders.

8.3 Stage C: every stable weak-order prefix

The current `PrefixRadiusArcTopologyFilter` only obtains a useful compiled propagation system when it has stable outer arcs and stable internal interfaces. There is an earlier exact condition that does not require an exterior seed.

C1. New `PrefixLengthFeasibilityFilter`. As soon as an interface path is stable under all descendants of the prefix, emit its homogeneous length row. Maintain the current strict-positive system incrementally.

Recommended subcascade:

1. same-sign row contradiction, already present in `LengthFeasibilityOracle`;
2. exact incremental row reduction and cache key canonicalization;
3. only if necessary, the tiny strict-positive LP, using the existing floating screen and exact rational certificate policy.

This test should run even when no exterior arc is stable yet.

C2. Period-length obstruction. For a rotation-fixed center tile, whenever a connected central interface becomes stable, run the row-space implication test from Theorem 7.1. This is cheaper than radius propagation and stronger than the old cyclic-hub pattern match because the length equality may be forced indirectly by other interface equations.

C3. Existing prefix radius-R topology. Only after the cheap length tests survive, run `PrefixRadiusArcTopologyFilter`. Its conservative behavior is correct: omitted unstable paths can only lose pruning, never invalidate a survivor.

8.4 Stage D: complete placement, before Nielsen–Levi

D1. Make the linear curriculum a production cascade. The current code already knows the nested sequence of metric systems but exposes `metric_curriculum()` primarily for diagnostics. In production, solve the nested systems in order and stop at the first exact failure.

The order should initially remain close to the current curriculum:

$$\text{positive lengths} \to \text{radius local} \to \text{smooth-turn local} \to \text{radius global} \to \text{smooth-turn global} \to \text{radius concavity} \to \text{isoperimetric},$$

with point turns as the separate angle system.

Whether this is faster than one final LP is an empirical question: several small LPs are beneficial only if early stages reject often enough. The solver already records calls, cache hits and elapsed time in the relevant oracles, so the decision can be made from data rather than intuition.

D2. Reuse row spaces and affine reductions. Nested systems repeatedly add constraints to almost the same matrix. Cache the reduced equality space and carry it forward instead of rebuilding exact elimination for every curriculum stage. This is an engineering optimization, not a change in mathematical semantics.

8.5 Stage E: before or inside generic word solving

E1. Preserve the complete Nielsen–Levi branching theorem. The current structural note correctly records that equality branches and exact-letter boundary cases are necessary for completeness. Any optimization must preserve the reconstruction bijections and mirror-equivariant canonicalization.

E2. Add specialized handlers for recognizable periodic SCCs. The current note also gives the counterexample $XY = YX$: two concurrent returning Nielsen branches generate a nontrivial finite-state path language, so compressing every return into independent or nested unary powers is not complete.

The recommended response is not to force a more complicated ad hoc loop compiler. Recognize standard periodic subproblems and switch representation. The commutation equation, one-variable residual systems, forced powers from rotational symmetry, and mirror-fixed words are the first candidates. Recompression is a mature exact alternative representation for word equations and can be used selectively rather than replacing the whole solver [6].

8.6 Stage F: decorated profile and geometric solver

No new hard rejection in this note replaces the existing final geometric stage. The purpose of the preceding filters is to ensure that far fewer candidates reach it.

9 Filter scheduling and telemetry

The pipeline should be ordered by measured conditional value, not by conceptual elegance.

Suppose two exact filters A and B have costs c_A, c_B and, on the population reaching them, rejection probabilities p_A, p_B. Running A then B has expected cost

$$c_A + (1 - p_A)c_B,$$

whereas B then A costs

$$c_B + (1 - p_B)c_A.$$

Thus A should precede B precisely when

$$\frac{c_A}{p_A} \leq \frac{c_B}{p_B},$$

when both rejection probabilities are positive. In a changing search population these quantities should be estimated conditionally at each stage.

The practical recommendation is to log the following quantities for every filter, separately by tile count, symmetry type, and center/noncenter branch:

- calls and cache hits;
- wall-clock and exact-certificate time;
- number of certified rejections;
- number of descendants avoided when a prefix is cut;
- first-failure stage in the linear curriculum;
- unresolved/truncated counts for conservative topology filters.

The objective is not simply rejection rate but *time saved downstream per unit filter time.*

10 Heuristics should remain separate from certificates

Several useful quantities should influence ordering without becoming formal rejection criteria:

- the strict positive-length margin;
- the perimeter-regularization margin if the LP version is retained;
- the number of unresolved radius-R images;
- the stage reached in the nested linear curriculum;
- the approximate monohedrality score from the inverse geometric optimizer discussed separately.

A candidate that is feasible only with a very small interface length may be less attractive computationally, but it is not impossible. The exact/heuristic boundary should remain explicit in the API and telemetry.

11 Research directions not yet ready for production rejection

11.1 Refining the relative profile by conditioning on the outer-arc length

Theorem 4.1 resolves the first Dido-type question that was open in the previous version of this note: without conditioning on the amount of outer contact, the minimum internal boundary length at area $\pi R^2/n$ is known exactly from the relative isoperimetric profile of the disk and is already production-ready after rational certification.

A stronger problem remains open and could prune much more. If a particular tile is already known to use an outer-circle arc of length a, determine the sharp lower bound

$$I_i \geq F_n(a)$$

for its internal boundary length at fixed area $\pi R^2/n$. The unconstrained relative minimizer chooses its own outer contact, whereas the solver frequently knows or linearly constrains a through symmetry and contour-length equations. A sharp conditional profile F_n would therefore add a nonlinear but one-dimensional cut to the perimeter-flow relaxation.

This conditional profile should not be used for rejection until its admissible class and extremizers are proved and its numerical evaluation is wrapped in directed bounds. It is now a genuine *refinement* of an established filter, rather than the only available area-versus-boundary idea.

11.2 Farkas-dual graph certificates for perimeter flow

Theorem 3.1 already gives a complete total-support characterization, so no LP is mathematically necessary at graph level. Nevertheless, the dual of the strict perimeter LP may produce particularly small obstruction witnesses adapted to the map representation. Independent-set Hall certificates are the first examples. It is worth cataloguing which dual extreme rays correspond to simple separators, bipartite deficits, or support edges excluded from all spanning cycle forests.

11.3 Multiple-orbit periodic curvature charges

Theorem 7.1 handles one connected interface whose length is an integral number of contour periods. A broader abelianized theorem may exist for several interface orbits whose total signed charge closes over one or more periods even when no individual interface has period length. This is the natural next generalization of the current cyclic-hub word certificate.

12 What should not be prioritized

The earlier graph-only attempts based on inradius, diameter, circumradius or convex-hull size should not be treated as the main exact-filter program for arbitrary Jordan tiles. Without an independent regularity or thickness hypothesis, arbitrarily thin corridors and complicated concavity leave these quantities too unconstrained. They may still be useful as ranking features after a concrete geometric realization or under explicit fatness hypotheses, but they are qualitatively weaker than conserved quantities such as perimeter, curvature, orbit structure and cyclic order.

Similarly, a high contact degree by itself is not an exact obstruction for arbitrary topological disks. Any theorem aimed at a high-degree center must exploit additional structure: a closed contact pattern, imposed symmetry, contour periodicity, exterior-circle data, or a regularity hypothesis.

13 Recommended implementation order

Priority	Change	Reason	Status
P0	Perimeter regularizability at map level	Complete graph-only necessary condition; total support/perfect matching; no contour enumeration required.	New theorem
P0	Stein rotation-orbit filter	Deletes complete symmetry strata; at 18 tiles all nontrivial rotations disappear, at 19 only orders 2,3,6,9 remain.	New theorem

Priority	Change	Reason	Status
P0	Prefix positive-length filter independent of outer seeds	Moves existing length necessity into the DFS as soon as stable rows exist.	Engineering
P0	Period-length center-interface test	Tiny exact row-space implication; strict generalization of the cyclic-hub obstruction.	New theorem
P1	Exterior-arc coincidence-class enumeration	Uses nondegenerate repetition and disjoint-or-coincident structure before general weak orders.	New strengthening + engineering
P1	Symmetry quotient of fixed-tile contour	Avoids enumerating repeated sectors independently.	Engineering
P1	Productionize nested linear curriculum	Stop on the cheapest exact infeasible prefix; benchmark with existing telemetry.	Engineering
P2	Incremental row-space / affine-state reuse	Reduces repeated exact elimination across nested systems and prefixes.	Engineering
P2	Specialized periodic word-SCC handlers	Avoids enormous Nielsen path languages for recognizable equations such as commutation.	Engineering / literature
R	Dido area–boundary inequality integrated into perimeter flow	Would add genuine disk geometry before words, but needs a certified derivation.	Open
R	Multi-orbit additive curvature certificates	Potential generalization of cyclic-hub obstruction beyond single period-length interfaces.	Open

14 Research status

At this point the search has produced four additions that are strong enough to justify implementation rather than further speculative exploration, plus one important source correction.

The source correction is non-negotiable: no stage may infer that every tile reaches the outer circle. Known monohedral examples contain tiles with empty exterior intersection. The safe statement used here is only that a *nonempty* exterior intersection is one connected circular arc. The exterior-arc repetition and rotation arguments have been audited under this convention and do not require universal boundary contact.

First, the perimeter condition is no longer merely "run another LP." It has a clean graph-theoretic identity: the loop-augmented contact support must be positively regularizable, equivalently every support edge must lie in a spanning cycle forest. This is exact, graph-only, polynomial, and available before offsets. Hall-type independent-set failures are cheap visible subcertificates.

Second, the relative isoperimetric profile of the disk gives a new exact map-level strengthening of that relaxation. Every tile has area $\pi R^2/n$, hence every tile must spend at least λ_n times the ambient circumference on internal interfaces. Because the ambient circumference is itself the sum of exterior-interface variables, these are linear inequalities on the existing perimeter-flow variables. With certified rational lower bounds for λ_n, the filter has the same conservative

exact-rejection semantics as the current rational LP infrastructure.

Third, rotational Stein branches admit a substantial whole-family rejection. The new exterior-arc bound forces at least two full peripheral orbits and therefore $n \geq 1 + 2q$. This is strong enough to delete entire symmetry strata in the solver's current size ranges.

Fourth, the cyclic-hub curvature argument extends directly to any connected center interface whose length system forces an integral number of rotational contour periods. Because the existing implementation already maintains exact length row spaces, the algorithmic cost of this theorem is very small.

The most immediate engineering opportunity is orthogonal to these theorems: expose positive-length feasibility and the nested linear curriculum much earlier in the weak-order DFS, and quotient occurrence enumeration by actual geometric symmetries rather than merely using symmetry as a graph label.

The next mathematical work should concentrate on two directions. The first is the *conditional* Dido profile at fixed exterior-arc length, now viewed as a strengthening of the established relative-isoperimetric filter rather than an unproved replacement for it. The second is a general additive-charge theorem for multiple periodic interface orbits. By contrast, unrestricted graph-only diameter/inradius arguments should be deprioritized unless the solver deliberately introduces a shape-regularity hypothesis.

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